Phase 5 - Project Development

Project: ComicCraft - AI Comic Story Creator **Team ID / members:** Fill in your team's details

Solution summary

ComicCraft is implemented in Python with FastAPI. Jinja2 renders the browser pages; Pydantic validates generation requests; separate services orchestrate the story, images, panel layout, and PDF export. Gemini and Hugging Face integrations are optional and configured through environment variables. Local fallback behavior supports a credential-free demonstration.

Feature inventory

Feature	Module / route	Status based on source review
Story prompt form	`templates/index.html`, `POST /generate`	Implemented
Request validation	`app/schemas.py`	Implemented
Five-panel story orchestration	`app/services/comic_service.py`	Implemented
Gemini story adapter	`app/services/gemini_service.py`	Implemented; requires valid key for hosted path
Hugging Face image adapter	`app/services/image_service.py`	Implemented; requires valid token for hosted path
Local illustration fallback	`app/services/image_service.py`	Implemented for demo behavior
Panel assembly	`app/services/layout_builder.py`	Implemented
Comic preview	`templates/comic_preview.html`	Implemented
PDF export and download	`app/services/pdf_service.py`, `/download/{filename}`	Implemented
JSON endpoint	`POST /generate-comic/json`	Implemented
Health and image diagnostics	`/health`, `POST /test-image`	Implemented

"Implemented" describes source-code availability. Verify runtime behavior on the machine used for the final presentation and fill in the test results in Phase 6.

Code organization and reuse

Quality check	Status	Evidence / note
Logical folder structure	Yes	`app`, `templates`, `static`, configuration and requirements are separated
Descriptive names	Yes	Service and route names describe their purpose
Modular design	Yes	Story, image, layout, and PDF work are separate services
Request validation	Yes	Pydantic request/response models
Error handling	Partial	Routes log server details and return user-facing errors; validate provider failures in live tests
Reusable modules	Yes	AI adapters and PDF/layout services are independently callable
Automated test suite	Not confirmed	Add and run tests before claiming automated coverage